6CS030: Big Data

Workshop - 3

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Exercise 2.1

1. Compare how you added the above data and how it differs from INSERT records in a relational database.

Following four are the major differences in between Relational Database and MongoDb:

1. Schema Flexibility

- **Relational:** We have to define a strict table structure before adding any data, and every single row must follow that exact template.
  - *Example:* INSERT INTO dept (deptno, dname) VALUES (20, 'RESEARCH'); will fail if the loc column isn't already in the table schema.
- **MongoDB:** We can just insert a document with any fields that we want, even if they're different from the other documents in the same collection.
  - *Example:* I can run db.dept.insertOne({ _id: 20, dname: "RESEARCH", budget: 5000 }) even if no other department has a "budget" field.

2. Table vs. Collection Creation

- **Relational:** The database will throw an error if we try to insert data into a table that hasn't been created yet using a CREATE TABLE command.
  - *Example:* Running an INSERT on a non-existent projects table results in a "table does not exist" error.
- **MongoDB:** If the collection doesn't exist, Mongo just creates it automatically the moment you run your first insertOne.
  - *Example:* Running db.new_dept.insertOne({ deptno: 40 }) works instantly, even if new_dept never existed before.

3. Data Formatting

- **Relational:** Data is stored in flat rows and columns, meaning we often have to split related info into different tables.
    - *Example:* To store a department's address, we might need a separate locations table and a foreign key.
- **MongoDB:** It uses BSON (like JSON), which allows you to nest arrays or other objects directly inside a single document.
    - *Example:* You can store the location as an object like loc: { city: "DALLAS", state: "TX" } right inside the dept document.

4. Primary Keys

- **Relational:** We usually set an INT column to AUTO_INCREMENT so the database manages the IDs itself.
    - *Example:* The ID is typically generated as 1, 2, 3, and so on, automatically upon insertion.
- **MongoDB:** Every document must have an _id, and if we don't provide one yourself, the database generates a unique ObjectId for you.
    - *Example:* If I leave out the _id, Mongo adds _id: ObjectId("69bca...") to the document by default.

2. Add Department 30, which has the following key/values: _id: 30, deptno: 30, dname: SALES and loc: CHICAGO

```
week3> db.dept.insertOne({_id:30, deptno:30, dname:"SALES", loc:"CHICAGO"})
{ acknowledged: true, insertedId: 30 }
week3> db.dept.find()
[
  { _id: 10, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 20, deptno: 20, dname: 'RESEARCH', loc: 'DALLAS' },
  { _id: 40, deptno: 40, dname: 'OPERATIONS', loc: 'BOSTON' },
  { _id: 50, deptno: 40, dname: 'OPERATIONS V2', loc: 'BOSTON' },
  { _id: 30, deptno: 30, dname: 'SALES', loc: 'CHICAGO' }
]
week3>
```

Exercise 2.2

1. Add the employees for Department 30.

Remember this time, some employees will have a commission (COMM):

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7499	ALLEN	SALESMAN	7698	20-FEB-95	1600	300	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7654	MARTIN	SALESMAN	7698	28-SEP-93	1250	1400	30
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7521	WARD	SALESMAN	7698	22-FEB-94	1250	500	30

```
week3> db.emp.insertMany([ { empno:7499,ename:"ALLEN",job:"SALESMAN",mgr:769
8,hiredate:new Date('1995-02-20'),sal:1600,comm:300,deptno:30}, {empno:7698,
 ename:"BLAKE",job:"MANAGER",mgr:7839,hiredate: new Date('1981-05-01'),sal:2
850,deptno:30}, {empno:7900,ename:"JAMES",job:"CLERK",mgr:7698,hiredate:new
Date('1981-12-03'),sal:950,deptno:30}, {empno:7654,ename:"MARTIN",job:"SALES
MAN",mgr:7698,hiredate:new Date('1993-09-28'),sal:1250,comm:1400,deptno:30},
 {empno:7844,ename:"TURNER",job:"SALESMAN",mgr:7698,hiredate:new Date('1981-
09-08'),sal:1500,comm:0,deptno:30},
| {empno:7521,ename:"WARD",job:"SALESMAN",mgr:7698,hirdate:new Date('1994-02
-22'),sal:1250,comm:500,deptno:30}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69bcb175ec032b5789ae5836'),
    '1': ObjectId('69bcb175ec032b5789ae5837'),
    '2': ObjectId('69bcb175ec032b5789ae5838'),
    '3': ObjectId('69bcb175ec032b5789ae5839'),
    '4': ObjectId('69bcb175ec032b5789ae583a'),
    '5': ObjectId('69bcb175ec032b5789ae583b')
  }
}
week3>
```

Exercise 2.4

· 2.4.1 Update the name of department 40 to: COMPUTING

```
week3> db.dept.find({deptno:40})
[
  { _id: 40, deptno: 40, dname: 'OPERATIONS', loc: 'WOLVERHAMPTON' },
  { _id: 50, deptno: 40, dname: 'OPERATIONS V2', loc: 'BOSTON' }
]
week3> db.dept.update({'deptno':40},{$set:{dname:'COMPUTING'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
week3> db.dept.find({deptno:40})
[
  { _id: 40, deptno: 40, dname: 'COMPUTING', loc: 'WOLVERHAMPTON' },
  { _id: 50, deptno: 40, dname: 'OPERATIONS V2', loc: 'BOSTON' }
]
week3>
```

· 2.4.1 Update the salary of employee number 7788 in department 20 to 3500

```
week3> db.emp.find({empno:7788})
[
  {
    _id: ObjectId('69bcaf16ec032b5789ae5835'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3100,
    deptno: 20
  }
]
week3> db.emp.update({'empno':7788,'deptno':20},{$set: {sal:3500}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
week3> db.emp.find({empno:7788})
[
  {
    _id: ObjectId('69bcaf16ec032b5789ae5835'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3500,
    deptno: 20
  }
]
week3>
```